



E-COMMERCE SALES ANALYTICS

A Power BI Dashboard Deep Dive: Revenue, Volume & Customer Behavior

Project Overview & Objectives

Why we built this dashboard, and what it needed to answer.

Overview

This project analyzes transactional e-commerce data to uncover revenue drivers, product performance, and customer purchasing patterns. A single-page Power BI dashboard consolidates the key metrics and visuals a business stakeholder needs to make fast, data-backed decisions.

Objectives

- Quantify total sales, order volume, and average order value across the full dataset.
- Track how sales trend year over year and identify seasonal or daily purchase patterns.
- Compare performance across product categories to find top and underperforming segments.
- Understand the customer base through age and gender breakdowns of spend.
- Give stakeholders self-service filtering by purchase date for ad-hoc analysis.



Built in Power BI Desktop

Interactive report connecting to the `ecommerce_sales_data` table — Category, Customer Age, Customer Gender, Order ID, Price, Product Name, Purchase Date/Time, and Quantity.

Key Performance Indicators

The three headline metrics from the dashboard's summary cards.



302.79M

Sum of Total Sales

Total revenue generated across every recorded transaction in the dataset.



549K

Sum of Quantity

Total number of units sold across all product categories.



3.03K

Average of Total Sales

Average transaction value — a proxy for average order size.

Power BI Dashboard Highlights

The full interactive report, built around five core visuals plus a date filter.



Sales by year & category

Bar and column visuals break total sales down by year and by product category.



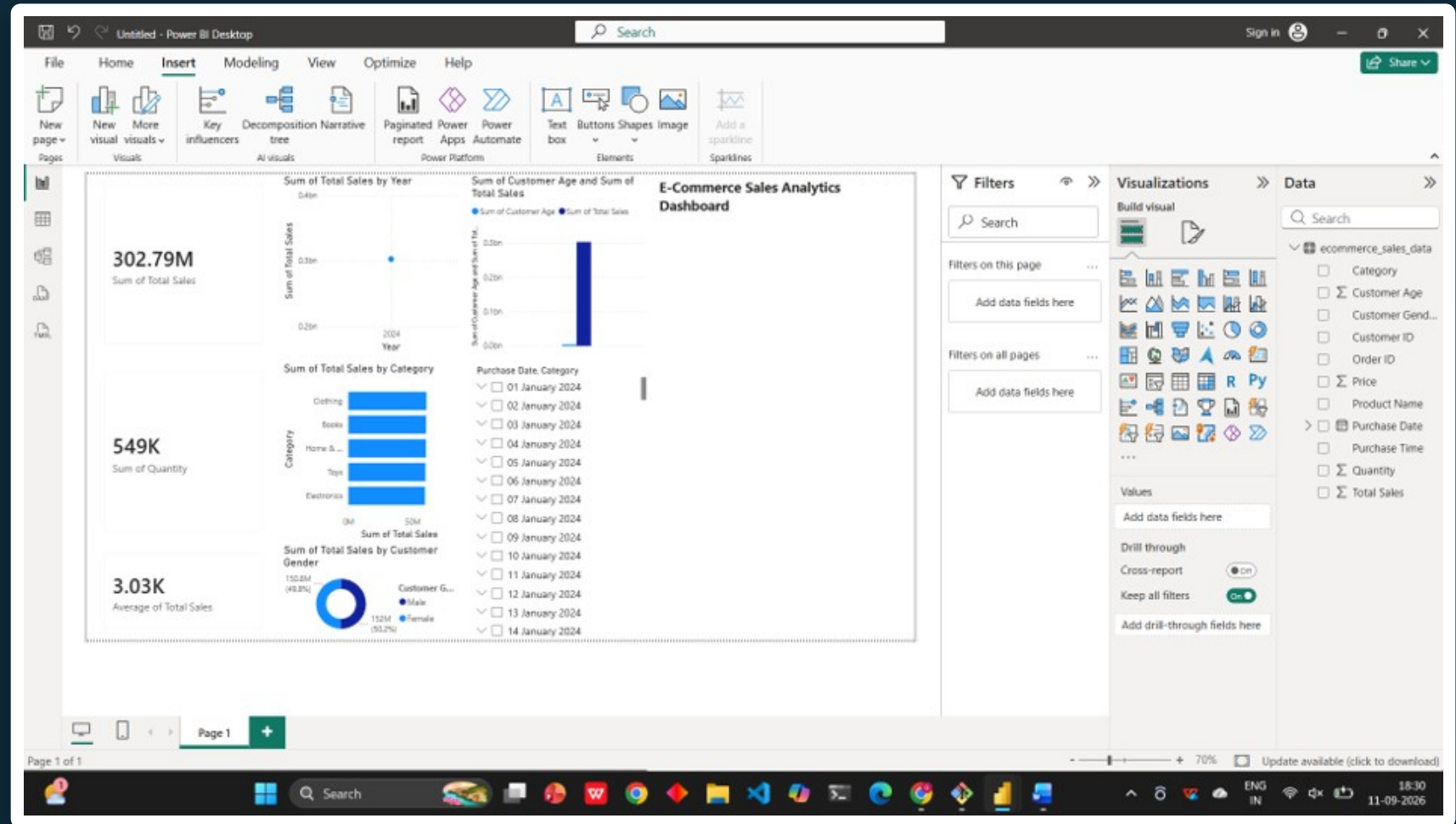
Customer age & gender

Scatter and donut visuals reveal how spend splits across age groups and gender.



Purchase-date drill-through

A day-level filter list lets viewers slice every visual down to a single date.





Business Insights & Recommendations

What the Data Shows

- Total revenue of 302.79M across ~549K units sold reflects a broad, steady transaction base rather than a few large orders.
- An average sale of ~3.03K per transaction suggests a mid-to-premium price point — worth validating against category-level margins.
- Category-level sales are uneven: a handful of categories (e.g. Clothing, Electronics) account for a disproportionate share of revenue.
- Customer age and gender splits show spend is not evenly distributed — some segments over-index on total sales relative to their size.

Recommended Next Steps

- Double down on top-performing categories with targeted promotions and inventory prioritization ahead of peak periods.
- Investigate underperforming categories for pricing, assortment, or visibility issues before cutting them.
- Build a customer segment strategy around the highest-value age/gender cohorts identified in the dashboard.
- Extend the report with month-over-month trend lines and cohort retention once more historical data is available.